

Giung Lee

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Education

Rice University, George R. Brown School of Engineering and Computing

- *Bachelor of Science in Computer Science, Minor in Statistics*
- Anticipated Graduation Date: May 2027
- Cumulative GPA: 3.77/4.0

Research Interests

Robot Safety, Safe Learning-Based Robotics, Risk-Aware Planning, Visual Navigation, Robot Manipulation, and Multi-Robot Systems

Publications & Manuscripts

Under Review

- **Giung Lee**, Weihang Guo, and Lydia E. Kavraki. *Foundation-Model-Guided Topology-Aware Semantic Risk Fields for Manipulation*. Under review, ICRA 2027.
- Geonyeong Ko[†], **Giung Lee**[†], and Changjoo Nam. *SAFER-Nav: Enhancing Safety for Visual Robot Navigation via Segmentation-Aware Fine-Tuning*. [arXiv] [Project Page] Under review, ICRA 2027.

[†] Equal contribution.

Research Experience

Kavraki Lab, Rice University – Houston, TX

Jan 2026 – Present

Undergraduate Researcher | Advisor: Prof. Lydia E. Kavraki

- Developed a **3D semantic risk field for robot manipulation** to represent physically feasible motions that are **collision-free but unsafe in context**.
- Designed an **object-conditioned, topology-aware 3D voxel field** whose risk distribution changes with the manipulated object. It combines foundation-model-derived directional risk priors with 3D scene geometry, geodesic spatial decay, and **topology-aware shielding based on the Euclidean-to-geodesic distance ratio**.
- **Led framework development and evaluation**, including GPU-based field construction, risk-aware manipulation planning, systematic counterfactual and component ablations, quantitative analysis, figure design, and primary manuscript drafting.
- Evaluated risk-aware trajectories against independent human-authored reference risk fields, reducing **mean path risk by 74.1% on average** across three household manipulation scenarios relative to collision-only planning under identical geometric constraints; further validated topology-aware shielding across vertical, lateral, and multi-surface barriers.

AI Robotics Lab, Sogang University – Seoul, South Korea

Jun 2025 – May 2026

Undergraduate Researcher | Advisor: Prof. Changjoo Nam

- Developed a **segmentation-aware approach to visual robot navigation** that trains the policy to **represent obstacles and traversable free space**, rather than relying solely on post-hoc trajectory correction or geometric safety priors.
- Designed a **segmentation-aware fine-tuning architecture** that incorporates traversability features into a frozen pretrained RGB navigation backbone through a **residual attention bridge and an auxiliary segmentation-only action branch**, allowing the policy to predict collision-aware waypoints while preserving goal-conditioned navigation.
- **Initiated the project by defining the research problem** and helping establish segmentation as an internal safety representation; co-led real-world experiments and contributed extensively to manuscript drafting and figure design.
- Evaluated SAFER-Nav across **three robot platforms and two unseen indoor environments**, including static- and dynamic-obstacle tests, against ViNT, NoMaD, and their CARE-augmented variants.
- Across static-obstacle tests, achieved **90%–100% goal arrival** with **0–0.4 collisions per run**, outperforming ViNT, NoMaD, and their CARE-augmented variants in goal arrival and collision frequency; across 20 dynamic-obstacle trials, SAFER-Nav remained **collision-free in all 20 trials**, compared with collisions in 14 trials for each of ViNT and NoMaD and 4–5 trials for the CARE-augmented baselines.

Robotics Lab, Texas A&M University – Remote
Undergraduate Researcher | Advisor: Prof. Yiwei Lyu

Jun 2026 – Present

- Developed a **safety intervention framework for frozen Vision-Language-Action (VLA) policies** that introduces geometry-derived physical safety references into the policy's generative process without fine-tuning the pretrained model.
- Designed a **reverse-flow latent steering method** that maps physical safety references into the VLA's latent flow source space and regenerates policy-native actions through the original frozen model.
- Developing an **RRT-based geometric recovery planner** for VLA safety dead-ends, generating collision-free recovery motions toward higher-clearance states to support safe handoff back to the frozen policy.

Work Experience

Teaching Assistant, COMP 450: Algorithmic and AI-Enabled Robotics | Department of
Computer Science, Rice University – Houston, TX

Aug 2026 – Present

- Grade homework for a 50-student robotics course covering configuration spaces, sampling-based planning, kinodynamics, and motion planning under uncertainty.
- Assess proof-based solutions for completeness and logical soundness, verifying that required cases and assumptions are addressed and the reasoning is internally consistent.

Teaching Assistant, STAT 310: Probability and Statistics | Department of Statistics, Rice
University – Houston, TX

Aug 2026 – Present

- Grade homework for a 98-student Probability and Statistics course, assessing statistical reasoning and problem-solving.

Korean Language Consultant | Department of Transnational Asian Studies, Rice University –
Houston, TX

Aug 2026 – Present

- Coach students in Korean pronunciation, conversation, and quiz review through four 30-minute one-on-one sessions each week.
- Evaluate students' Korean reading and pronunciation during oral exams, scoring performance for course assessments.

Teaching Assistant, COMP 215: Introduction to Program Design | Department of
Computer Science, Rice University – Houston, TX

Aug 2025 – May 2026

- Mentored 120+ students in Java, object-oriented programming, data structures, and software design principles.
- Held weekly office hours and review sessions; graded programming assignments using standardized rubrics and delivered constructive feedback.

UAS Mission Commander (Sergeant) | U.S. Army (KATUSA), Engineering Battalion – Camp
Hovey, South Korea

Nov 2022 – May 2024

- Served as a squad leader in the UAS platoon, leading soldiers during daily duty operations, morning physical training, and platoon-level coordination.
- Supervised 150+ UAV flight operations as UAS Mission Commander, coordinating pre-flight safety checks, mission execution, and operational readiness.
- Acted as a military liaison between the U.S. Army and the ROK Army, supporting safe UAV flight training in Korean airspace and compliance with both ROK and U.S. military aviation standards.

Honors & Awards

- **Korea Presidential Science Scholarship** | 1 of 20 students selected nationwide
- **Rice University President's Honor Roll** | Spring 2026, Fall 2025
- **1st Place** | 2024 KT&G Alliance Students' Venture Forum
- **Creativity Award** | 2024 Army Start-up Business Competition

Projects

SPACE-K | UAV Tracking and Communication System

Mar 2023 – Apr 2024

- Led a student team in a business competition project to develop a low-altitude tracking and communication system, replacing LTE-based communication with SATCOM.
- Developed the technical deployment scenario, business model, and prototype, earning 1st Place at the 2024 KT&G Alliance Students' Venture Forum and a Creativity Award at the Army Start-up Business Competition.

Service

Korea Connect | Rice Alzheimer's Buddies – Houston, TX

Nov 2024 – Present

- Lead small-group sessions that bring Rice students and Korean older adults together through bilingual companionship and group activities aimed at reducing social isolation.
- Facilitate Korean-English communication and help organize cultural and social activities each semester in coordination with students, participants, and community partners.

Relevant Coursework

- **Robotics & Machine Learning:** Algorithmic and AI-Enabled Robotics, Introduction to Modern Robotics, Practical Machine Learning, Reasoning About Algorithms
- **Mathematics & Statistics:** Matrix Analysis for Data Science, Multivariable Calculus, Applied Multivariate Statistics, Probability and Statistics

Technical Skills

- **Programming:** Python, C/C++, Java, R
- **Robotics & Machine Learning:** PyTorch, JAX, ROS 2, MuJoCo, PyRoki, OpenCV, NumPy
- **Tools:** Git/GitHub, Docker, Conda, Linux (Ubuntu), LaTeX
- **Languages:** Korean, English